



Education

Introduction to Cloud Foundry (LFD132x)

Deploying and managing applications securely and efficiently is critical in today's technology organizations. As a result, organizations are turning to Kubernetes to run their workloads, only to find massive complexity associated with the platform. Learn about Cloud Foundry, the modern model to deliver cloud-native applications to any infrastructure, including Kubernetes, in this free online course.

Course Overview

This course explains what Cloud Foundry is and why it should be used. The course starts by highlighting the main use cases for the Cloud Foundry platform, and then discusses the relationship between Cloud Foundry and Kubernetes before providing an immersive, hands-on experience that can be completed by anyone. The course also explores the features and capabilities of Cloud Foundry, highlighting the impact from three perspectives: developers, operations, and security and compliance teams.

This course prepares you to make informed decisions regarding development platform selection and use in custom software organizations. It helps you understand the needs of custom software organizations from the standpoint of developers, operators, and security and compliance teams. In addition, this course highlights the challenges of developing and operating custom software applications throughout their lifecycle, and offers a perspective on what this process can look like for teams that select the right platform.

Course Instructor(s)



Steve Greenberg is the Founder and CEO of Resilient Scale (<https://rscale.io>), a consultancy focused on building capabilities through collaboration, training, and mentoring. He is a longtime application developer and engineer with over twenty years of experience. Steve has been working with Cloud Foundry since 2012 and has been a contributor since 2014. He lives in Colorado in the United States and spends his free time mountain biking, fly fishing, and making things in his workshop.

Audience

While aimed at a technical audience, this course is also widely applicable to non-technical audiences. It is not necessary to be a developer or operator to complete this course. The hands-on portion does not require any special skills, software installs, or environments, as it is designed for a broad range of backgrounds.

Prerequisites

- Web browser

Course Length

- 12 hours

Course Learning Objectives

By the end of this course, you should be able to:

- Understand the significant capabilities of the Cloud Foundry platform and the value it brings to custom software development teams
- Understand the security and compliance implications of using a developer-focused platform like Cloud Foundry
- Articulate the difference between Cloud Foundry and infrastructure platforms like Kubernetes
- Understand the open source model that governs the Cloud Foundry project

Course Outline

- Welcome!
- Chapter 1. Introduction to Cloud Foundry
In this section, we delve into Cloud Foundry's role in technology organizations. First, we cover Cloud Foundry at a high level, the guiding principles, and the core concerns the platform aims to address. We then discuss the relationship between Cloud Foundry and Kubernetes, highlighting how Cloud Foundry can unlock the complexity of Kubernetes before getting hands-on with the platform.
- Chapter 2. Developer Concerns
In this section, we look at the capabilities of Cloud Foundry from the perspective of a typical developer. We look at application lifecycle management, supported workloads, application containerization, team constructs, networking and routing, services, and visibility and awareness.
- Chapter 3. About the project
In this section, we discuss the structure and engagement model of the open-source project. We cover the open-source model and ecosystem, contributors and member companies, and discuss engaging with the project and community.
- Final Exam (Verified Track Only)

edX Platform

If you are using edX for the first time, we strongly encourage you to start by taking a free 'how to use edX' course that the team at edX has made available. In this course, you will learn how to navigate the edX platform, how to connect with other edX learners, how to answer problems on the edX platform, how grades work in edX courses, and how to complete your first course.

Click [here](#) to register for “*DemoX*” and you will be on your way. You will find the edX platform simple and intuitive.

Getting Help

For any **technical issues** with the edX platform (including login problems and issues with the Verified Certificate), please use the **Help** icon located on the upper right side of your screen.

One great way to interact with peers taking this course and resolving any **content-related issues** is via the **Discussion Forums**. These forums can be used in the following ways:

- To discuss concepts, tools, and technologies presented in this course, or related to the topics discussed in the course material.
- To ask questions about course content.
- To share resources and ideas related to AI, ethical frameworks, and more.

We strongly encourage you not only to ask questions, but to share with your peers opinions about the course content, as well as valuable related resources. The Discussion Forums will be reviewed periodically by The Linux Foundation staff, but it is primarily a community resource, not an 'ask the instructor' service.

Course Timing

This course is entirely self-paced; there is no fixed schedule for going through the material. You can go through the course at your own pace, and you will always be returned to exactly where you left off when you come back to start a new session. However, we still suggest you avoid long breaks in between periods of work, as learning will be faster and content retention improved.

The chapters in the course have been designed to build on one another. It is probably best to work through them in sequence; if you skip or only skim some chapters quickly, you may find there are topics being discussed you have not been exposed to yet. But this is all self-paced and you can always go back, so you can thread your own path through the material.

Learning Aids

Besides simple exposition through text and figures, this course uses several additional methods to present and solidify the learning material, including videos, scenarios, external resources, and knowledge check questions (Verified Certificate track only).

Audit and Verified Tracks

You can enroll into an audit or a verified track. In an audit track, you will have access to all ungraded course content: course readings, videos, and learning aids, but no certificates are awarded when auditing. You will not be able to access any graded content (knowledge check questions at the end of each chapter, and the final exam).

In order to receive a certificate, you will need to obtain a passing grade (please refer to the "Grading" section below), verify your identity with edX, and pay a fee. Once all edX requirements have been met, you can download your certificate from the Progress tab.

To learn more about audit and verified tracks, visit [edX Help Center > Certificates](#).

Grading (Verified Certificate track only)

At the end of each chapter, you will have a set of graded **knowledge check questions**, that are meant to further check your understanding of the material presented. The grades obtained by answering these knowledge check questions will represent **20%** of your final grade.

The remaining **80%** of your final grade is represented by the score obtained in the **final exam**. The final exam is located at the end of the course and it consists of 18 questions.

You will have a maximum of two attempts to answer each knowledge check and final exam question (other than True/False questions, in which case, you have only one attempt). You are free to reference your notes, screens from the course, etc., and there is no time limit on how long you can spend on a question. You can always skip a question and come back to it later.

In order to complete this course with a passing grade, you must obtain a passing score (knowledge check and final exam) of minimum 70%.

Course Progress and Completion (Verified Certificate track only)

Once you complete the course (including knowledge check questions and final exam), you will want to know if you have passed. You will be able to see your completion status using the **Progress** tab at the top of your screen, which will clearly indicate whether or not you have achieved a passing score.

About The Linux Foundation

About The Linux Foundation

[The Linux Foundation](#) provides a neutral, trusted hub for developers to code, manage, and scale open technology projects. Founded in 2000, The Linux Foundation is supported by more than 1,000 members and is the world's leading home for collaboration on open source software, open standards, open data and open hardware. The Linux Foundation's methodology focuses on leveraging best practices and addressing the needs of contributors, users and solution providers to create sustainable models for open collaboration.

The Linux Foundation hosts Linux, the world's largest and most pervasive open source software project in history. It is also home to Linux creator Linus Torvalds and lead maintainer Greg

Kroah-Hartman. The success of Linux has catalyzed growth in the open source community, demonstrating the commercial efficacy of open source and inspiring countless new projects across all industries and levels of the technology stack.

As a result, the Linux Foundation today hosts far more than Linux; it is the umbrella for many critical open source projects that power corporations today, spanning virtually all industry sectors. Some of the technologies we focus on include big data and analytics, networking, embedded systems and IoT, web tools, cloud computing, edge computing, automotive, security, blockchain, and many more.

The Linux Foundation Events

The Linux Foundation hosts an increasing number of events each year, including:

- Open Source Summit North America, Europe, Japan and China
- Embedded Linux Conference + OpenIoT Summit North America and Europe
- Open Source Leadership Summit
- Open Networking Summit North America and Europe
- KubeCon + CloudNativeCon North America, Europe and China
- Automotive Linux Summit
- KVM Forum
- Linux Storage Filesystem and Memory Management Summit
- Linux Security Summit North America and Europe
- Cloud Foundry Summit
- Hyperledger Global Forum
- And many more.

To learn more about The Linux Foundation events and to register, click [here](#).

The Linux Foundation Training

The Linux Foundation offers several types of training:

- Classroom
- Online
- On-site
- Events-based.

To get more information about specific courses offered by The Linux Foundation, click [here](#).

The Linux Foundation Certifications

The Linux Foundation certifications give you a way to differentiate yourself in a job market that's hungry for your skills. We've taken a new, innovative approach to open source certification that

allows you to showcase your skills in a way that other peers will respect and employers will trust:

- You can take your certification from any computer, anywhere, at any time
- The certification exams are either performance-based or multiple choice
- The exams are distribution-flexible
- The exams are up-to-date, testing knowledge and skills that actually matter in today's IT environment.

For a list of currently offered certifications, click [here](#).

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